



# Bridging Standards and Deployments

At the intersection of Media and Connectivity, for generations to come

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5G-MAG Chair

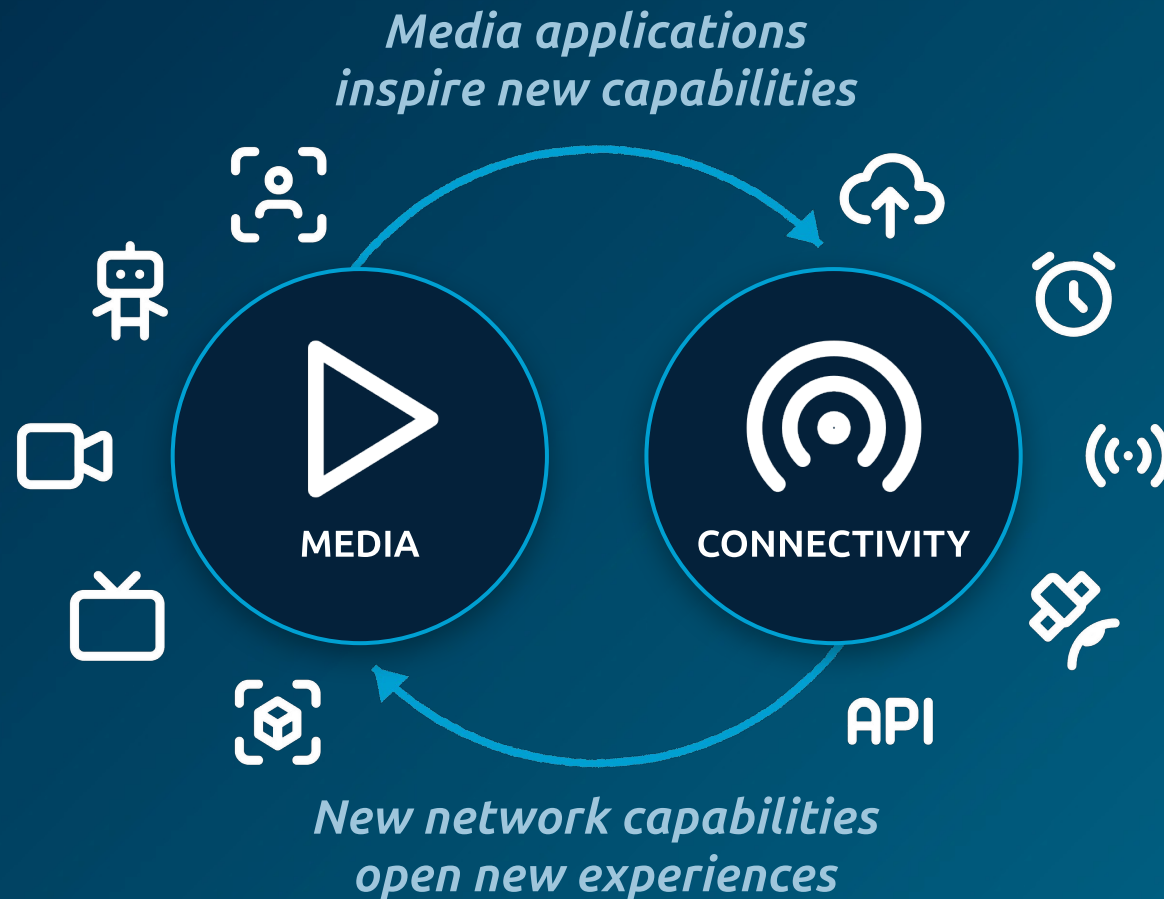


**STREAMING – PRODUCTION – BROADCAST – REAL-TIME COMMS – AVATAR – EXTENDED REALITY  
5G/6G – NON-TERRESTRIAL NETWORKS – DATA COLLECTION – NETWORK APIs – COMPUTING – AI  
CONNECTED DEVICES – VIRTUAL WORLD – VOLUMETRIC VIDEO – CODECS – PROTOCOLS – AUDIO**



# Media and Connectivity

-  **Fast tech evolution**
-  **Global ecosystem**
-  **Built better together**
-  **Solution to pain-points**



Connected Devices  
Streaming Platforms  
Next Generation Codecs  
Internet-based Protocols  
eXtended Reality  
Wireless Video  
Real-Time Comms  
Data and Analytics  
Content Production  
Satellite  
Network APIs  
Artificial Intelligence  
Quality of Service  
Latency  
Energy Consumption  
Synchronization  
Spatial Media

# Some examples of our current areas of work

The full track at [www.5g-mag.com/tech](http://www.5g-mag.com/tech)

Don't see your topic here?  
[Bring your topic or project →](#)



## Content Delivery and Streaming

5G Media Streaming (5GMS)  
Content Delivery Protocols  
UE Data Collection, Reporting and Event Exposure



## Point-to-Multipoint Communication

5G Multicast Broadcast Services (MBS)



## Connected Media Production

CAMARA Connectivity Quality Management APIs  
Non-Public Networks  
Time-Sensitive Communications (TSC)



## Real-Time Communications (RTC)

Real-time Media Communication (RTC) Architecture



## Non-Terrestrial Networks (NTN)

Non-Terrestrial Networks in 5G Systems



## Immersive Media

Conversational Avatar Communication with MPEG ARF  
MPEG V3C Immersive Platform  
XR/3D Scenes with MPEG-I Scene Description  
Dynamic Mesh Coding



## Towards 6G Media

AI/ML Evaluation Framework  
Towards 6G Media  
AI Traffic Characterization



## Broadcast

DVB-I Services over 5G Systems  
5G Broadcast - TV and Radio Services  
5G Broadcast - Emergency Alerts



## Testbeds & Evaluation Frameworks

Beyond 2D Video Experiences

# From requirements to products

SPECIFICATIONS DRIVING  
MEDIA + CONNECTIVITY

IMPLEMENTED AND  
PUSHED INTO ACTION



# Technology & Blueprints

Understand it. Prove its value. Scale it.



Specification analysis for  
real-world applications



Making standards  
actionable



Lowering the barrier to  
entry



## Technical Analysis

Understanding how  
specifications support  
application requirements.



## Implementation Blueprints

Turn standards into  
implementation-ready  
blueprints.

## Examples of **Standards Development Organizations (SDO)**



# Standardisation Activities

Advancing standardization with implementation and experience-based feedback.



Active engagement with standards bodies



Requirements based on application needs



Identification of bugs and enhancements



## Requirements towards SDOs

Requirements towards standards bodies, backed by industry inputs and surveys.



## Industry Surveys

Share your input to help shape the requirements 5G-MAG brings to standards bodies.



## Feedback to SDOs

Per-SDO feedback tracked on GitHub, and how an issue is carried through to a correction or liaison input.



## Liaison Statements & Inputs

Tables of LS sent to and received from SDOs.



## Workshops for Standards

The workshops that fed directly into 3GPP study and work items.

# Standardisation activities in numbers

Our work at a glance

182

issues raised on specifications through GitHub

3GPP

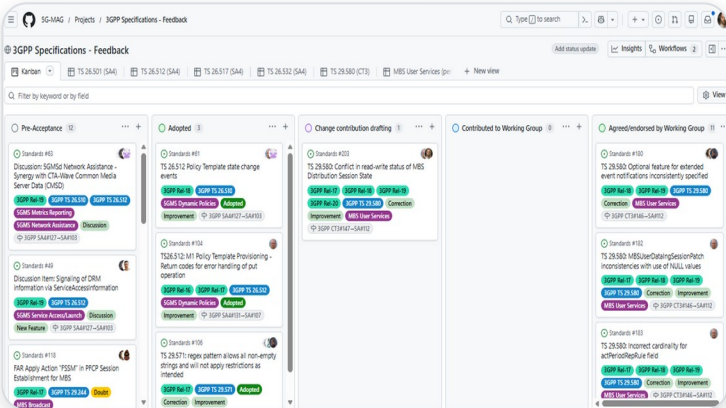
engagement as Market Representation Partner

35

inputs and liaison statements to SDOs

6

presentations at 3GPP requirements workshops



Issues found on specifications through GitHub



Profiles of Standards Specifications



Requirements towards Standards Bodies

# Media Connectivity Software Accelerator

Our work at a glance



Early feedback and validation



Software projects & Dev community



No duplication and fast deployment



Open by design and IPR-friendly



## Reference Tools

Standards, turned into open code anyone can build on.



## Testbeds & Evaluation Frameworks

Reproducible test environments and benchmark frameworks.



## Application Showcases

Use-case driven implementations towards real-world applications.

# Enabling open-source software from the start of every project

TODAY: OPEN SOURCE AFTER THE FREEZE

WITH OPEN SOURCE FROM THE START

Evaluation

**Duplicated prototypes.** Each organization builds its own to evaluate a draft; results differ and feedback is slow.

**One shared baseline.** Drafts are validated against running code before the freeze, while gaps are cheap to fix.

Interoperability

**Late interop bugs.** Testing starts from scratch at plugfests; errors, some in the specification, surface late.

**Faster plugfests.** Implementers start from the same code: shorter preparation, faster root-cause analysis.

Deployment

**Slow deployment.** Years can pass from a frozen specification to production-ready code; early adopters carry the cost alone.

**Earlier deployment.** End-to-end demos and trials run before commercial products, and feed evidence back into the standard.

5G-MAG  
IN PRACTICE

182  
issues raised on specifications  
through GitHub

35  
inputs and liaison statements to SDOs

Reference Tools  
And testbeds supporting prototypes,  
reference implementations and  
technology evaluation

# Make open-source software and evaluation platforms work-plan deliverables, not optional extras

Work item launch

Drafting

Freeze and deployment

1

## Plan it at launch

Define the open-source software and evaluation platform track from day one: where the code lives, its scope, the evaluation metrics. Reuse existing open platforms.

2

## Engage implementers during drafting

Involve open-source communities while the specification is written, and invest in that development. Running code surfaces issues that text review cannot.

3

## Make the value tangible

End-to-end demos and benchmark platforms show that the standard works. They lower cost for the whole ecosystem: members and non-members alike, start-ups, academia.

IN ONE LINE, FOR EVERY NEW WORK ITEM

“Open-source software and evaluation platform: home and scope identified at launch.”

# Software Accelerator in numbers

The full track at [5g-mag.com/developer](https://5g-mag.com/developer)

22

projects across media  
and connectivity



56

project repositories,  
public and early access



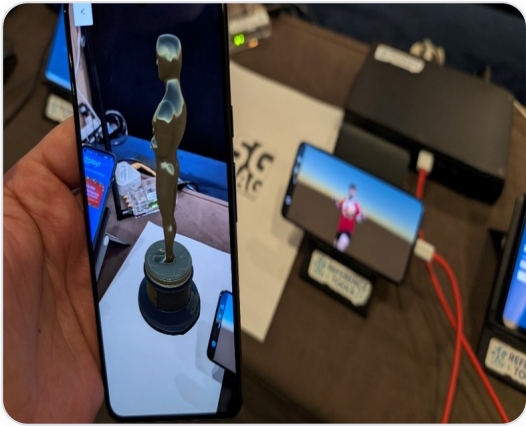
18,577

clones: developers  
pulling the code



Monthly

Dev Public Call and  
rolling releases



# Growing community of developers

Organizations contributing to the Media Connectivity Software Accelerator.



BBC



Big Blue Marble



Bitstem



Dolby Laboratories



EBU



Ericsson



Fraunhofer FOKUS



Huawei



idTOLU



InterDigital



iTEAM



Motion Spell



Nokia



Philips



phine.tech



Qualcomm



Resillion



Sofia Digital



Universitat Politècnica de València

# A note on the Open-Source Software Licenses

Protecting IPR. Enabling industry collaboration.

## What is the License Model?

1. The 5G-MAG Public License v1.0 is a modified version of Apache 2.0 which allows:
  - **Free Non-Commercial Use:** freely use, copy, modify, and distribute the code for non-commercial purposes, including study, academic research, testing, and validation.
  - **Commercial Exploitation:** use the reference tools and software within commercial services, trials, and live deployments, but:
    - **FRAND Patent Licensing:** if used commercially, the built-in patent clause ensures that any essential contributor patents are made available under **FRAND** (Fair, Reasonable, and Non-Discriminatory) terms.
2. Repositories with software derived from other licenses keep their own license according to their own terms.

## How to Contribute?

1. Sign an Individual or Corporate Contributor License Agreement (CLA).
2. Anybody can become a contributor (no need to be a 5G-MAG member to contribute code).
3. 5G-MAG members get priority in support from the 5G-MAG Project Office, and additional development resources may be provided. 5G-MAG members define priorities and where to dedicate resources.
4. Membership of 5G-MAG remains open to the industry, to boost and scale your software project by joining forces.



# Validation, Interop Plugfests, Demos and Applications

Specifications to code. Code to deployments.



Make value proposition tangible to the industry



Interoperability through early testing and plugfests



Supporting application development and trials



## End-to-End Demos and Use Cases

User-facing demonstrations that abstract technical complexity.



## Interoperability Events and Plugfests

With shared code, so interop does not start from scratch at plugfests.



## Showcase at Industry Events and Trials

Promotion and demonstration of value, not just specs.

# Where to meet us

2 recurrent large events and a yearly conference, plus workshops and an open monthly call.



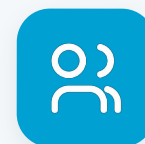
## MWC Barcelona

Recurrent large event in Barcelona, with a 5G-MAG stand.



## IBC

Recurrent large event in Amsterdam, with Reference Tools demos.



## Future Media Townhall

Yearly conference at IBC, co-organized by 5G-MAG and IEEE BTS.



## OSCAR Workshop

Open-Source Core, Applications and RAN.



## OSMART Workshops

Open-Source Media Application Reference Tools.



## Dev Public Call

Open monthly session, no membership needed. Last Friday of the month, 13:00 CET/CEST.



**Join, collaborate, sponsor software  
development & scale your projects**

[www.5g-mag.com/membership](http://www.5g-mag.com/membership)

JOIN THE EFFORT

# Addressing industry pain points with open specifications

## Member-driven

we add resources to  
their efforts

## Open

to the industry to  
engage, collaborate

## Agile

at the pace of  
standards and market

## Pragmatic

with actionable  
activities & outcomes

3

**WORKGROUPS** focused on **TECHNOLOGY AND STANDARDS,**  
**DEVELOPMENT AND ECOSYSTEM,** and **PROMOTION AND COMMUNICATION**

JOIN THE EFFORT

# What you get that you won't get alone

We offer a neutral framework to collaborate within minimal bureaucracy, focused on results.



## Shape standards early

Shape the standards and technologies before others set them without you.



## Early access to pre-public code

Access implementations before publication and be faster to market.



## Prepare for plugfests & interop

Test your implementation in advance and build reference vector traces.



## Mutualised effort to grow your project

From idea to reference code to demo and trial, on shared effort.



## Exposure at events, demos and trials

Build once, together, instead of funding the same work alone.



## De-risk on deployments

Back technologies that actually get deployed, not specs that stall.

JOIN THE EFFORT

# Our current membership



NIXENET

ASTRUM  
MOBILE

ATEME  
Captivate your audience

BBC

Belgian  
Tower Company

BIG BLUE  
MARBLE  
ORF

broadcast  
networks europe

Dolby

EBU  
OPERATING EUROVISION AND EURORADIO

EITOWERS

ERICSSON

esa

EUTELSAT

france•tv

Fraunhofer  
FOKUS

HAIVISION

HUAWEI

BTS  
IEEE Broadcast  
Technology  
Society

interdigital

kronehit

mbi

MCC

Media  
Broadcast

NOKIA

ONE  
Media  
Technologies

PHILIPS

PHINE-  
TECH

Qualcomm

Rai

GRUPPO MEDIASET  
RTI

rtve

ses

SMART  
MOBILE  
LABS

swisscom

SWISS TXT

SWR

TemX  
COMMUNICATIONS

TRedess

UNIVERSITAT  
POLITÈCNICA  
DE VALÈNCIA

XGN

Yotta  
Media

# Membership information and Fee Structure

An association, a platform to collaborate, and a specific reason to be for each part of the media and connectivity chain.

|                                          |                                                    |                                          |
|------------------------------------------|----------------------------------------------------|------------------------------------------|
| Membership Fee Category<br><b>LARGE</b>  | Annual Revenue<br><b>&gt; €1 billion</b>           | Annual Membership Fee<br><b>€ 15,000</b> |
| Membership Fee Category<br><b>MEDIUM</b> | Annual Revenue<br><b>€100 million – €1 billion</b> | Annual Membership Fee<br><b>€ 10,000</b> |
| Membership Fee Category<br><b>SMALL</b>  | Annual Revenue<br><b>€5 million – €100 million</b> | Annual Membership Fee<br><b>€ 5,000</b>  |
| Membership Fee Category<br><b>MICRO</b>  | Annual Revenue<br><b>&lt; €5 million</b>           | Annual Membership Fee<br><b>€ 2,000</b>  |

**Micro** also includes universities, regulators, public research bodies, institutions, NGOs, and non-profit organizations.

Membership information, sponsorship opportunities, partnerships, questions on fees?  
Contact **Eva Markvoort** at **markvoort@5g-mag.com**

JOIN THE EFFORT

# Get involved

Ways to work with us, member or not.



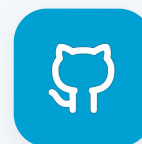
## Bring your topic or project

Don't see your topic here? Bring it, or your project, as a member.



## Early Access

Request access to Reference Tools and Testbeds still in development.



## Contribute code

No membership needed: sign the CLA and open a pull request on GitHub.



## Join the developers

The Dev Community on Slack and the Reference Tools mailing list.



## Stay informed

Email updates, the 5G-MAG Magazine and The Handshake podcast.



## Follow us

News on LinkedIn; talks and demos on YouTube.



# Join the Effort

Help shape what connected media looks like next and scale your projects with the industry.

[Become a member →](#)

[www.5g-mag.com/membership](http://www.5g-mag.com/membership)



GitHub



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